

Machine Learning Overview and Data Representation

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(Attendance Code: 839919)

In the last lecture,

- Module Information
- Contents of the module

Today's Content



Contents of the module (cont.)



What is machine learning?



A few applications of machine learning

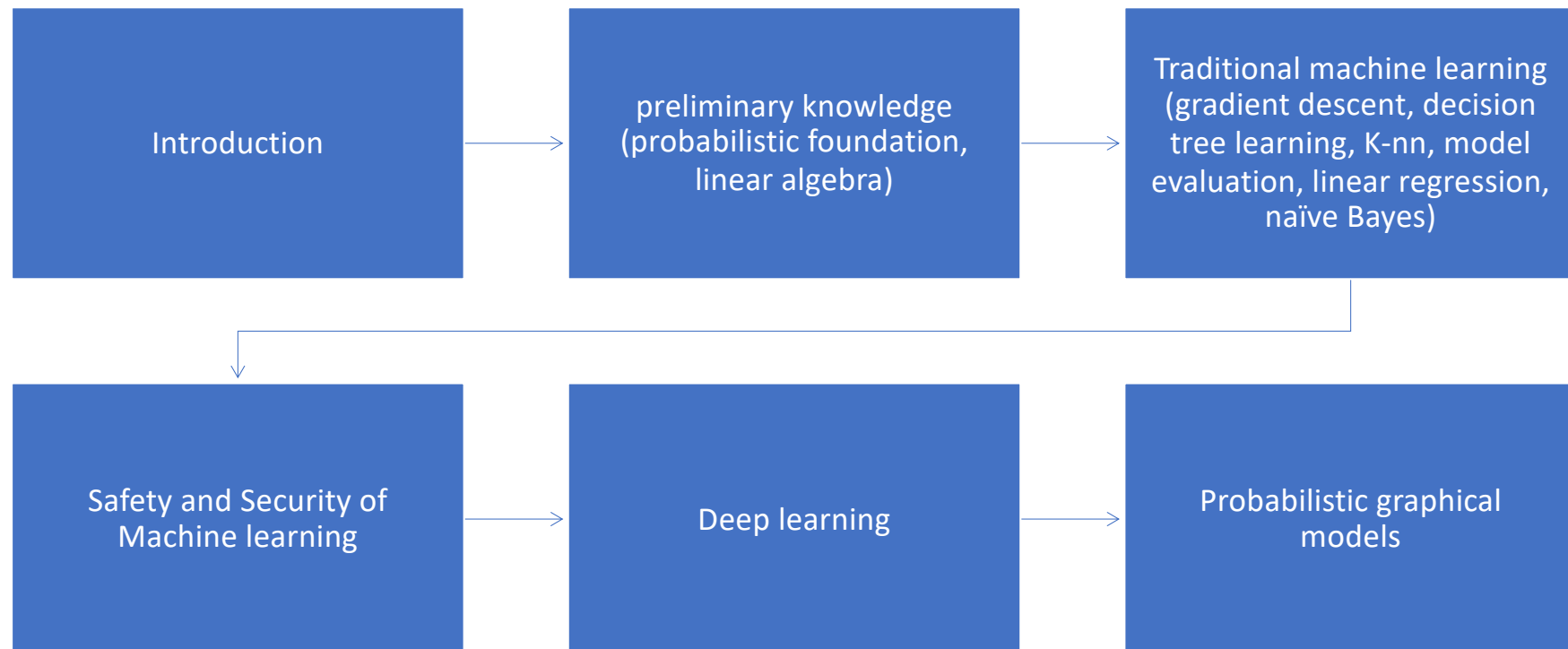


consider how to represent instances as fixed-length feature vectors



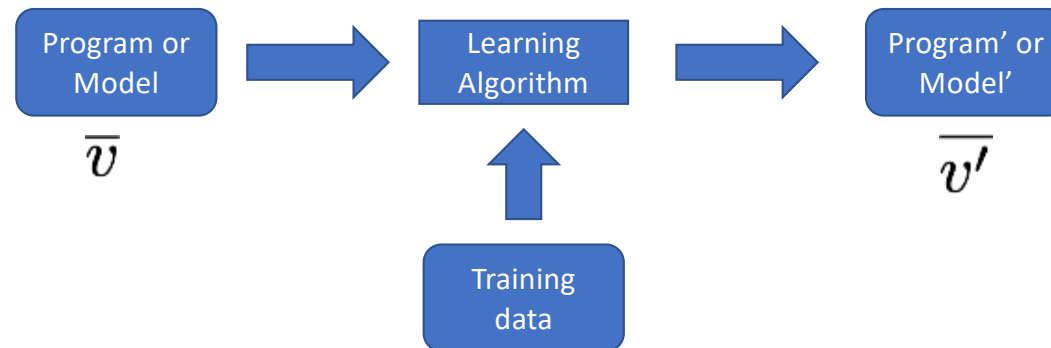
Data preprocessing

Contents of this module



What is Machine Learning?

- (Software) programs that can improve their performance by applying learning algorithm on training data
- Typically the program has a (large) number of parameters whose values are learnt from the data



Why learn instead of program?

- If the design of the agent can be improved, why wouldn't the designers just program in that improvement to begin with?
 - The designer cannot anticipate all possible situations that the agent might find itself in
 - The designers cannot anticipate all changes over time
 - Sometimes a human programmer has no idea how to program a solution

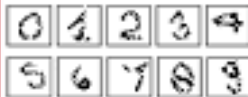


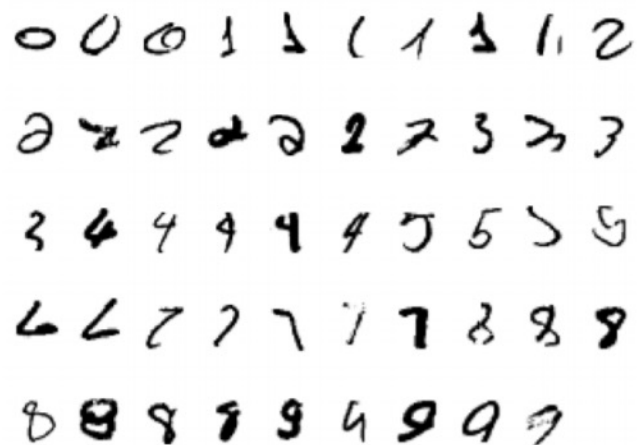
A few applications of machine learning

Where Machine Learning is used/useful?

- Can be applied in situations where it is very challenging (= impossible) to define rules by hand, e.g.:
 - Face detection
 - Speech recognition
 - Stock prediction
- When the application is able to be programmed with reasonable efforts, DO NOT use machine learning!

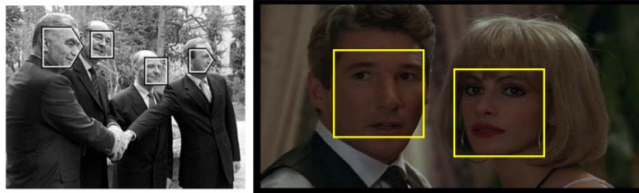
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0	10	16	110	238	255	244	245	243	250	249	255	222	103	10	0
0	14	170	255	255	244	254	256	253	245	255	249	253	251	124	1
2	95	255	228	255	251	254	211	141	116	122	215	251	238	255	45
13	217	243	255	135	33	226	52	2	0	10	13	232	255	255	36
16	229	252	254	49	12	0	0	7	7	0	70	237	252	235	62
6	41	245	255	212	25	11	9	3	0	115	236	243	255	37	0
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0	0	6	1	0	52	153	233	255	252	147	37	0	0	4	1
0	0	5	5	0	0	0	0	0	14	1	0	6	6	0	0

[illegible]
$$f : \mathbb{R}^{784} \rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$



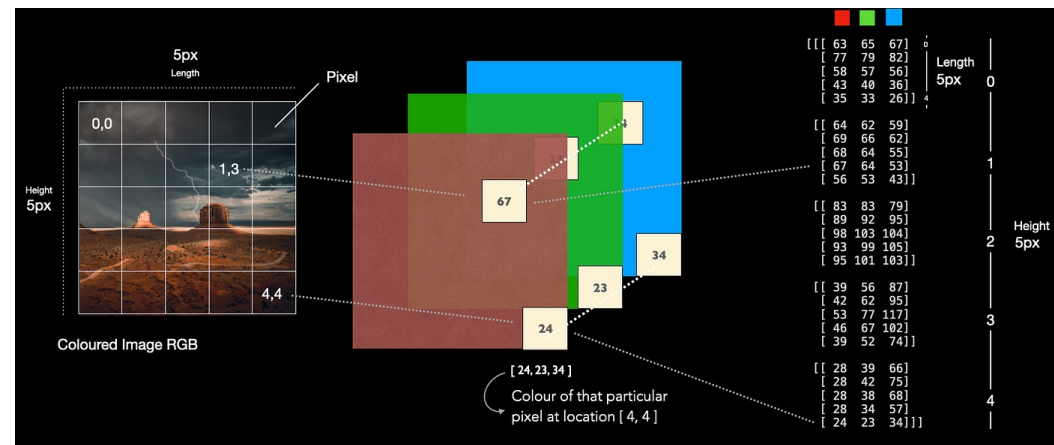
How to proceed ...

- As a supervised classification problem
 - Start with training data, e.g. 6000 examples of each digit
-
- Can achieve testing error of 0.4%
 - One of the first commercial and widely used ML systems (for zip codes & checks)



Example 2: Face detection

- Again, a supervised classification problem
- Need to classify an image window into three classes:
 - non-face
 - frontal-face
 - profile-face



RGB image digital representation



Classifier is learnt from labelled data

- Training data for frontal faces
 - 5000 faces
 - All near frontal
 - Age, race, gender, lighting
 - 10^8 non faces
 - faces are normalized
 - scale, translation (a **translation** is a geometric **transformation** that moves every point of a figure or a space by the same distance in a given direction)



- This is a classification problem
- Task is to classify email into spam/non-spam
- Data x_i is word count, e.g. of viagra, outperform, “you may be surprised to be contacted” ...
- Requires a learning system as “enemy” keeps innovating

Example 3: Spam detection

$\longleftrightarrow d = 5$

your	0.23	0.18	0.34	0.76	0.62
shirt	0.64	0.23	0.21	0.03	0.83
looks	0.98	0.59	0.76	0.65	0.45
nice	0.11	0.43	0.30	0.22	0.92

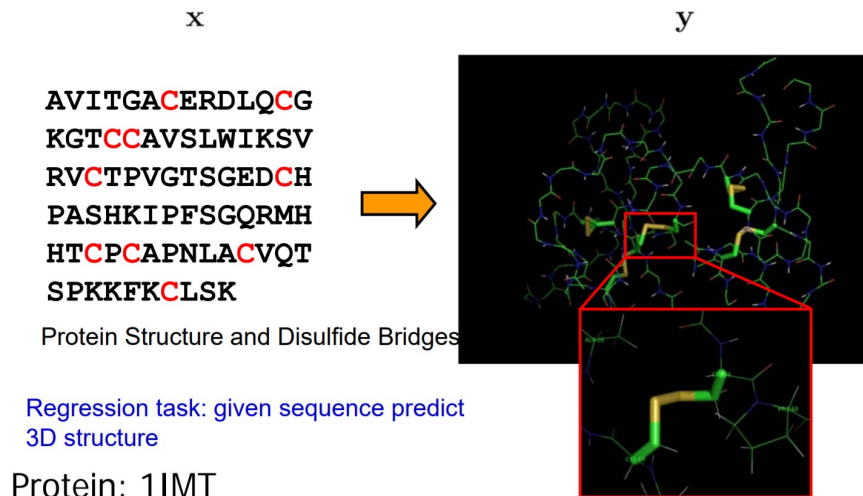


Example 4: Stock price prediction

- Task is to predict stock price at future date
- This is a regression task, as the output is continuous

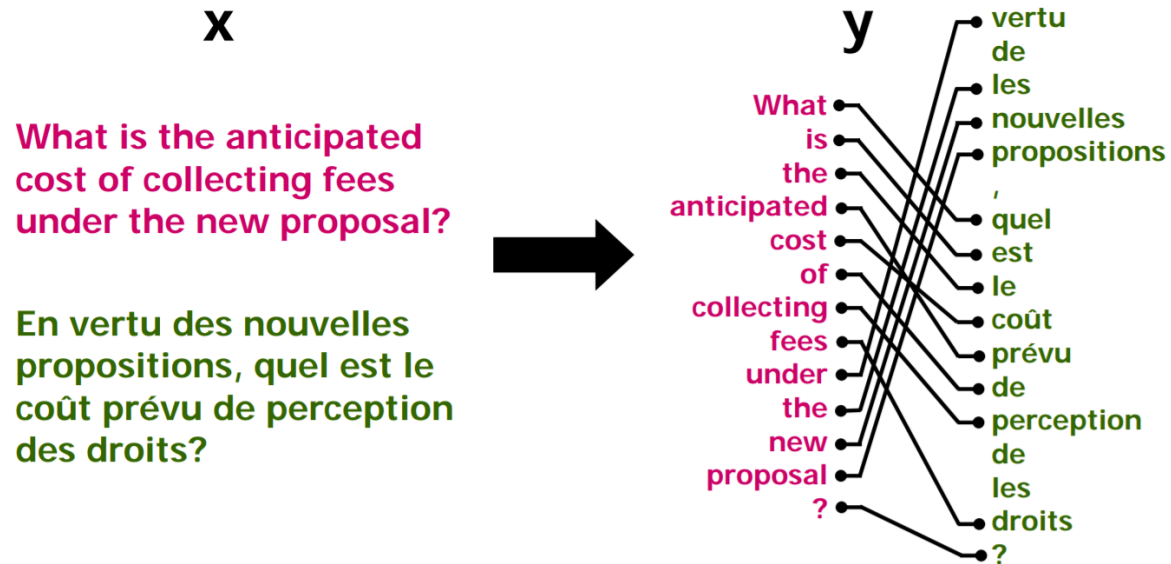
Example 5: Computational biology

- **Protein structure prediction** is the inference of the three-dimensional structure of a [protein](#) from its [amino acid](#) sequence
- based on the dataset alone, the algorithm can learn how to combine multiple [features](#) of the input data into a more abstract set of features from which to conduct further learning



Web examples: Machine translation

Use of **aligned** text



Web examples: Recommender systems

People who bought Hastie ...

Frequently Bought Together

Customers buy this book with [Pattern Recognition and Machine Learning \(Information Science and Statistics\) \(Information Science and Statistics\)](#) by Christopher M. Bishop



Price For Both: **£104.95**

Add both to Basket

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represent instances as fixed-length feature
vectors

Can I eat this mushroom?

- I don't know what type it is – I've never seen it before. Is it edible or poisonous?



Can I eat this mushroom?

suppose we're given examples of edible and poisonous mushrooms (we'll refer to these as *training examples* or *training instances*)

edible



poisonous

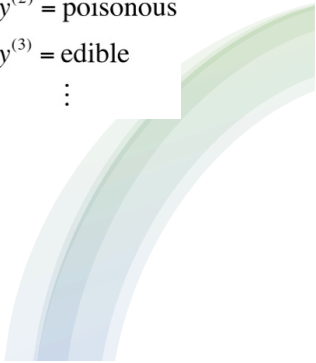




Representing instances using feature vectors

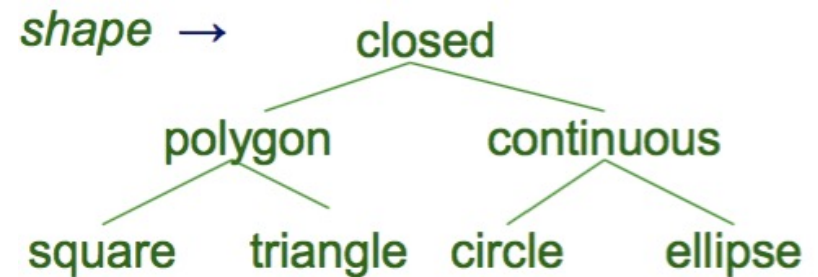
- we need some way to represent each instance
- one common way to do this: use a fixed-length vector to represent *features* (a.k.a. *attributes*) of each instance
- also represent *class label* of each instance

	cap-shape	cap-surface	cap-color	bruises	odor		class
$\mathbf{x}^{(1)}$	=	<bell,	fibrous,	gray,	false,	foul,...>	$y^{(1)}$ = edible
$\mathbf{x}^{(2)}$	=	<convex,	scaly,	purple,	false,	musty,...>	$y^{(2)}$ = poisonous
$\mathbf{x}^{(3)}$	=	<bell,	smooth,	red,	true,	musty,...>	$y^{(3)}$ = edible
			$\vdots$				$\vdots$



Standard feature types

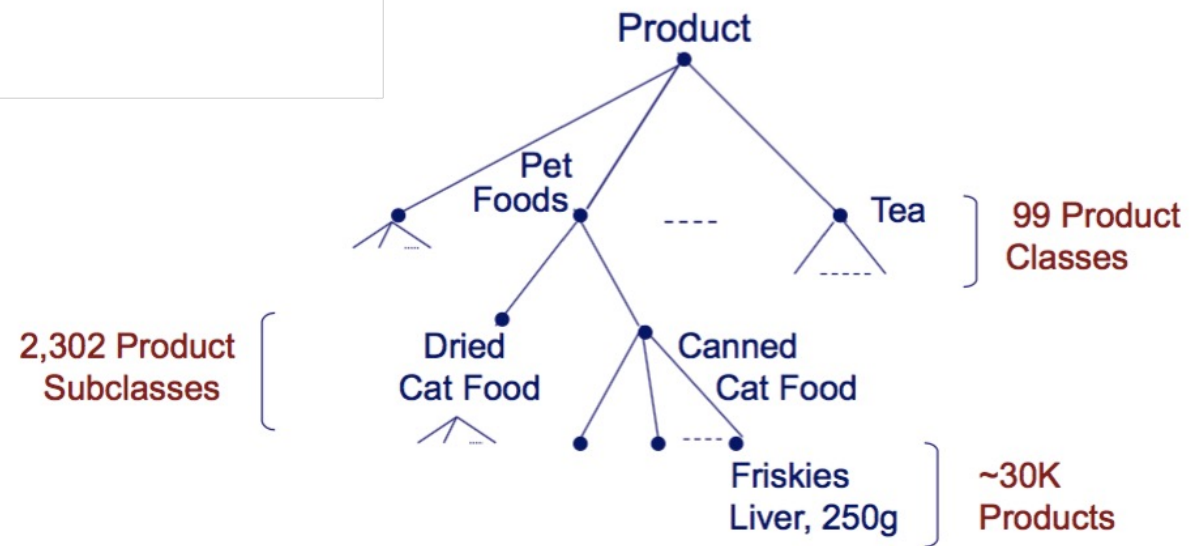
- *nominal* (including Boolean)
 - no ordering among possible values
 - e.g. $color \in \{red, blue, green\}$ (vs. $color = 1000$ Hertz)
- *ordinal*
 - possible values of the feature are totally ordered e.g. $size \in \{small, medium, large\}$
- *numeric (continuous)*
 - E.g., $weight \in [0...500]$
- *hierarchical*
 - possible values are partially ordered in a hierarchy



Feature hierarchy example

- Lawrence et al., *Data Mining and Knowledge Discovery* 5(1-2), 2001

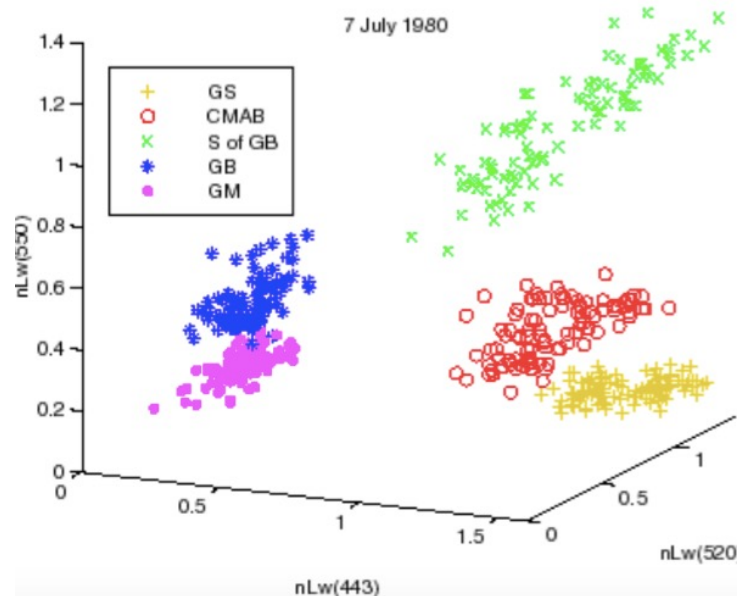
Structure of
one feature



Feature space

- we can think of each instance as representing a point in a d-dimensional feature space where d is the number of features

example: optical properties of oceans in three spectral bands
[Traykovski and Sosik, *Ocean Optics XIV Conference Proceedings*, 1998]



How about a 3-dimensional space for height, weight, and body fat percentage?

Another view of the feature-vector representation: a single database table

	feature 1	feature 2	...	feature d	class
instance 1	0.0	small		red	true
instance 2	9.3	medium		red	false
instance 3	8.2	small		blue	false
...					
instance n	5.7	medium		green	true

Data Preprocessing

10.4.3.1 Mean Normalization

removes the mean from each data sample, i.e.,

$$\mathbf{x}' = \mathbf{x} - \bar{\mathbf{x}} \quad (10.43)$$

where $\bar{\mathbf{x}} = \frac{1}{|D|} \sum_{\mathbf{x} \in D} \mathbf{x}$ is the mean of the dataset D .

10.4.3.2 Standardization or Normalization

requires, on top of the mean normalisation, all features to be on the same scale, i.e., every sample $\mathbf{x}$ is converted into

$$\mathbf{x}' = \frac{\mathbf{x} - \bar{\mathbf{x}}}{\sigma_D} \quad (10.44)$$

where σ_D is the standard deviation of the dataset D .

Data Preprocessing

10.4.3.3 Whitening

requires that the covariance matrix of the converted dataset is the identity matrix—1 in the diagonal and 0 for the other cells. It first applies the mean normalisation on the dataset D to get D' , and then apply a whitening matrix W on every sample, i.e., let $\mathbf{x}'' = \mathbf{W}\mathbf{x}'$, such that $\mathbf{W}\mathbf{W}^T = \Sigma^{-1}$ and Σ is the non-singular covariance matrix of D' .

Depending on what $\mathbf{W}$ is, we have Mahalanobis or ZCA whitening ($\mathbf{W} = \Sigma^{-1/2}$), Cholesky whitening ($\mathbf{W} = \mathbf{L}^T$ for $\mathbf{L}$ the Cholesky decomposition of Σ^{-1}), or PCA whitening ($\mathbf{W}$ is the eigen-system of Σ^{-1}).